UNIT - IV

IMPACT OF JETS & HYDRAULIC TURBINES

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

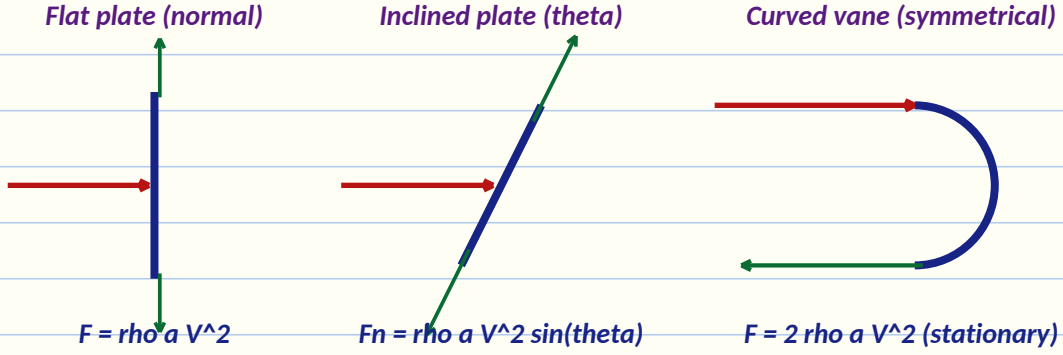
- Hydrodynamic force of jets on stationary and moving flat, inclined and curved vanes
- Velocity triangles at inlet and outlet; work done and efficiency
- Hydraulic turbines: classification; Pelton wheel and its design
- Francis turbine and its design; efficiency; draft tube theory
- Characteristic curves of hydraulic turbines; cavitation: causes and effects

1. Impact of Jets

1.1 Principle

A jet of water striking a vane exerts a **force** equal to the **rate of change of momentum** of the jet in that direction (Newton's second law): $F = \rho Q (V_{in} - V_{out})$.

- Jet velocity V (m/s), area $a = \pi d^2/4$, discharge $Q = a V$, mass flow rate $\rho a V$.



1.2 Stationary vanes

Vane	Force on vane (direction of jet)
Flat plate normal to jet	$F = \rho a V^2$
Inclined plate (angle θ to jet)	Normal force $Fn = \rho a V^2 \sin\theta$; $Fx = \rho a V^2 \sin^2\theta$; $Fy = \rho a V^2 \sin\theta \cos\theta$

Symmetrical curved vane (jet at centre, tip angle θ)	$F = \rho a V^2 (1 + \cos\theta)$; semi-circular vane ($\theta = 0$): $F = 2 \rho a V^2$
Unsymmetrical curved vane (jet at one tip, tip angles θ, φ)	$Fx = \rho a V^2 (\cos\theta + \cos\varphi)$; $Fy = \rho a V^2 (\sin\theta - \sin\varphi)$

- A curved vane gives **more force** than a flat plate because the jet is turned (deflected); a semi-circular vane gives twice the force.

1.3 Moving vanes (plates moving with velocity u in the direction of the jet)

- **Relative velocity** of jet = (V - u). Only the **relative** mass flow striking the single moving plate is $\rho a (V - u)$.

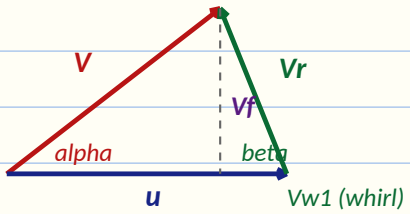
Case	Force	Work done / s	Efficiency
Single moving flat plate (normal)	$\rho a (V - u)^2$	$\rho a (V - u)^2 u$	$2u (V - u)^2 / V^3$; max 29.6% at $u = V/3$
Series of flat plates (wheel)	$\rho a V (V - u)$	$\rho a V (V - u) u$	$2u (V - u) / V^2$; max 50% at $u = V/2$
Single curved vane (moving)	$\rho a (V - u)^2 (1 + \cos\theta)$	F x u	$2u (V - u)^2 (1 + \cos\theta) / V^3$
Series of curved vanes (wheel)	$\rho a V (V - u)(1 + \cos\theta)$	F x u	$2u (V - u)(1 + \cos\theta) / V^2$; max $(1 + \cos\theta)/2$ at $u = V/2$

- **Series of vanes** (many vanes on a wheel) intercept the whole jet, so the entire mass flow $\rho a V$ is used (not $\rho a (V - u)$). For semi-circular vanes ($\theta = 0$) the maximum efficiency is 100%.
- Jet KE per second = $(1/2) \rho a V^3$ (input power). Efficiency = work done per second / KE per second.

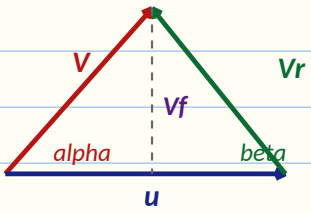
1.4 Velocity triangles

- At **inlet** and **outlet** of a moving vane draw: V (absolute velocity of jet), u (vane velocity), Vr (relative velocity), Vw (whirl component, along u), Vf (flow component, perpendicular to u), α (angle of V with u), β (angle of Vr).

Inlet triangle



Outlet triangle



$V = u + Vr$ (vector sum); $Vr1 = V1 - u1$ (for tangential entry); $Vr2 = k\ Vr1$ (k = friction factor, $k \le 1$)

Work done per second = $\rho Q (Vw1 \pm Vw2)\ u$ (Euler's equation, per unit mass: $u (Vw1 \pm Vw2)$)

- Use + sign when Vw2 is in the opposite direction to u (blade angle at outlet is acute / obtuse as per triangle), and - sign when Vw2 is in the same direction as u.

1.5 Solved Problems

Problem 1: Jet d = 50 mm, V = 20 m/s strikes (a) a stationary flat plate normally; (b) a plate moving at 8 m/s in the direction of the jet; (c) a series of plates at 10 m/s. Find force, power and efficiency.

- $a = 1.9635 \times 10^{-3}\ m^2$; jet KE rate = $0.5 \times 1000 \times 1.9635 \times 10^{-3} \times 20^3 = 7854$ W.

(a) $F = \rho a V^2 = 1000 \times 1.9635 \times 10^{-3} \times 400 = 785.4$ N

(b) $F = \rho a (V - u)^2 = 1.9635 \times 144 = 282.7$ N; power = $282.7 \times 8 = 2262$ W; $\eta = 2262 / 7854 = 28.8\%$

(c) $F = \rho a V (V - u) = 1.9635 \times 20 \times 10 = 392.7$ N; power = 3927 W; $\eta = 50\%$

Problem 2: Same jet on an inclined stationary plate at $\theta = 30°$ to the jet axis.

- $Fn = \rho a V^2 \sin\theta = 785.4 \times 0.5 = 392.7$ N; Fx (along jet) $= Fn \sin\theta = 196.4$ N; $Fy = Fn \cos\theta = 340.1$ N.

Problem 3: Same jet on a symmetrical curved vane (tip angle $\theta = 20°$): stationary and moving ($u = 5$ m/s, single vane).

- Stationary: $F = \rho a V^2 (1 + \cos 20°) = 785.4 \times 1.9397 = 1523.5$ N.
- Moving: $F = \rho a (V - u)^2 (1 + \cos 20°) = 1.9635 \times 225 \times 1.9397 = 857$ N; power $= 857 \times 5 = 4285$ W.

2. Hydraulic Turbines: Classification

2.1 Definition and heads

- A **hydraulic turbine** converts the hydraulic energy of water (head) into mechanical energy (shaft power), which is converted into electrical energy by a generator.
- **Gross head Hg** = difference of head race and tail race levels. **Net (effective) head H = Hg - hf** (hf = head loss in penstock).

2.2 Classification

Basis	Types
Action of water	**Impulse** (Pelton): only kinetic energy, atmospheric pressure at runner. **Reaction** (Francis, Kaplan): both kinetic and pressure energy; runner is full
Head	High (> 250 m): Pelton. Medium (60 to 250 m): Francis. Low (< 60 m): Kaplan, propeller
Direction of flow	Tangential (Pelton); radial inward (old Francis); mixed (modern Francis); axial (Kaplan, propeller)
Specific speed	Low (Pelton, Ns up to 60); medium (Francis, 60 to 400); high (Kaplan, > 300)
Shaft position	Horizontal or vertical

2.3 Impulse vs Reaction turbine

Basis	Impulse (Pelton)	Reaction (Francis / Kaplan)
Energy at inlet	Only kinetic (after the nozzle)	Kinetic + pressure
Pressure over runner	Atmospheric	Falls from inlet to outlet
Casing	Not essential (only to prevent splashing)	Essential, water-tight
Admission	Partial (jets)	Full circumference
Draft tube	Not required	Required
Head and discharge	High head, low discharge	Medium / low head, large discharge

2.4 Efficiencies and power

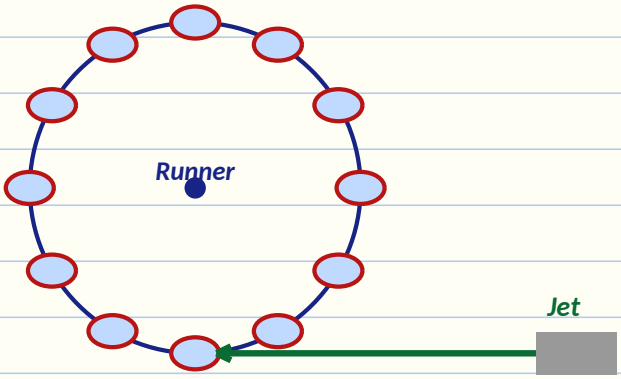
Water power WP = ρ g Q H ; Shaft power SP = 2π N T / 60

Hydraulic efficiency ηh = Runner power / Water power ; Mechanical efficiency ηm = Shaft power / Runner power

Volumetric efficiency ηv = (Q - leakage) / Q ; Overall efficiency ηo = SP / WP = ηh x ηm x ηv

3. Pelton Wheel (Impulse Turbine)

3.1 Main parts



Impulse turbine: high head, low discharge. Works at atmospheric pressure.

Bucket is double-hemispherical with a splitter in the centre.

- **Penstock** (pipe from reservoir); **Nozzle with spear** (needle valve regulates discharge); **Runner with buckets** (double hemispherical cups with splitter); **Casing** (prevents splash, guides water to tailrace); **Breaking jet** (stops the runner quickly after shut-down); **Deflector** (protects penstock from water hammer).

3.2 Working

- Water at high head is converted into a high-velocity jet in the nozzle; jet hits the splitter at the centre of the bucket, divides into two halves, and is deflected through about **165°** (not 180° to avoid striking the next bucket). Almost all KE is given to the wheel; water leaves with very small velocity.

3.3 Velocity triangle and work done

- Inlet: $V1$ = jet velocity (tangential), $u1 = u2 = u$, $Vr1 = V1 - u$, $Vw1 = V1$. Outlet: $Vr2 = k\,Vr1$, deflection angle $(180° - \varphi)$, $\varphi = 15°$, $Vw2 = Vr2 \cos\varphi - u$.

$$\text{Force} = \rho Q (V1 - u)(1 + k\cos\varphi) \;;\; \text{Power} = \rho Q (V1 - u)(1 + k\cos\varphi)\, u$$

$$\eta h = 2 (V1 - u)(1 + k\cos\varphi)\, u / V1^2 \;;\; \text{maximum at } u = V1/2:\ \eta h,max = (1 + k\cos\varphi)/2$$

- If $k = 1$ and $\varphi = 0$, $\eta h,max = 100\%$ (ideal). In practice $u \approx 0.45\ V1$ (speed ratio about 0.43 to 0.48) with $\eta 0 = 80$ to 90%.

3.4 Design of a Pelton wheel

Item	Formula / typical value
Jet velocity	$V1 = Cv\sqrt{(2\,g\,H)}$, $Cv = 0.97$ to 0.99
Bucket (wheel) velocity	$u = \varphi s\sqrt{(2\,g\,H)}$, speed ratio $\varphi s = 0.43$ to 0.48
Jet diameter	$Q = n\,(\pi/4)\,d^2\,V1$ (n = number of jets, up to 6)
Wheel (pitch circle) diameter D	$D = 60\,u / (\pi\,N)$; jet ratio $m = D/d = 10$ to 15 (up to 20)
Number of buckets	$Z = 15 + D/(2\,d) = 15 + m/2$ (typically 17 to 24)

Bucket dimensions	Width = 5 d, depth = 1.2 d, axial width about 3d to 4d
Specific speed	Ns = N √P / H^(5/4); 10 to 35 (one jet), up to 60 (multi-jet)
Regulation	Spear valve (main), jet deflector (quick closure)

3.5 Solved Problem 4 (Pelton design)

Net head 300 m, Cv = 0.98, speed ratio 0.45, jet diameter 100 mm, wheel diameter 1.4 m, deflection 165° (φ = 15°), k = 0.9. Find V1, u, Q, N, power, efficiencies, specific speed.

- V1 = 0.98 √(2 x 9.81 x 300) = 0.98 x 76.72 = **75.2 m/s** ; u = 0.45 x 76.72 = **34.5 m/s** .
- Q = (π/4)(0.1)² x 75.2 = **0.591 m³/s** ; N = 60 u / π D = 60 x 34.5 / (π x 1.4) = **471 rpm** ; D/d = 14, Z = 15 + 7 = 22 buckets.
- Vr1 = 75.2 - 34.5 = 40.7; Vr2 = 0.9 x 40.7 = 36.6; Vr2 cos 15° = 35.35.

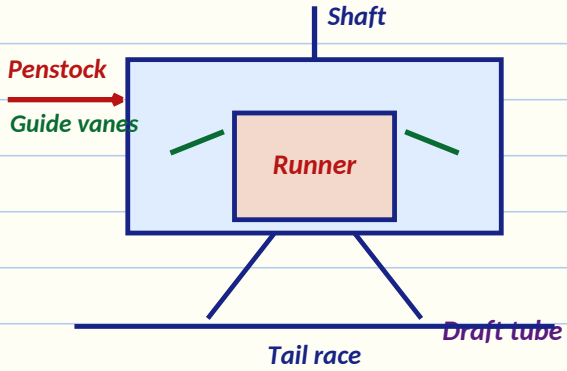
Power = 1000 x 0.591 x (40.7 + 35.35) x 34.5 = 1.55 x 10⁶ W = *1550 kW*

- Jet KE rate = 0.5 x 1000 x 0.591 x 75.2² = 1.67 MW, so **ηh = 1.55 / 1.67 = 92.8%** (based on jet KE).
- Water power = 1000 x 9.81 x 0.591 x 300 = 1.74 MW, so efficiency based on water power = 89.2% (includes nozzle loss).

Ns = N √P / H^1.25 = 471 x √1550 / 300^1.25 = 471 x 39.4 / 1249 = 14.9 (single-jet Pelton range)

4. Francis Turbine (Reaction, Mixed Flow)

4.1 Main parts



Reaction turbine (mixed flow):
medium head, medium discharge.
Water enters radially inward, leaves axially into draft tube.
Runner flows full of water.

- **Spiral (scroll) casing** with decreasing area (constant velocity); **stay ring / stay vanes** ; **guide vanes (wicket gates)** adjustable to regulate discharge and give correct entry angle; **runner** with 16 to 24 curved blades; **draft tube** (diverging) to the tailrace.

4.2 Working

- Water enters the casing at high pressure, passes through the stay and guide vanes, enters the runner **radially inwards**, gives energy to the blades while its pressure falls, and leaves **axially** (mixed flow) into the draft tube.

4.3 Velocity triangles and work done

- Inlet: V1, Vw1, Vf1 (radial), $u1 = \pi D1 N / 60$, inlet blade angle θ, guide vane angle α. Outlet: $u2 = \pi D2 N/60$, Vw2, Vf2.

Work done per second = ρ Q (Vw1 u1 ± Vw2 u2) ; Euler head He = (Vw1 u1 ± Vw2 u2) / g

For radial discharge (Vw2 = 0): P = ρ Q Vw1 u1 ; ηh = Vw1 u1 / (g H)

Flow rate Q = π D1 B1 Vf1 = π D2 B2 Vf2 (reduce by blade thickness factor if given)

- $\tan\theta = Vf1 / (Vw1 - u1)$ (for Vw1 > u1, θ < 90°); $\tan\alpha = Vf1 / Vw1$.

4.4 Design parameters

- Speed ratio **Ku = u1 / √(2gH) = 0.6 to 0.9** ; flow ratio **Kf = Vf1 / √(2gH) = 0.15 to 0.30** ; **B1/D1 = 0.1 to 0.4** ; D2 depends on specific speed (D2/D1 less than 1 for slow runners); **Ns = 60 to 300** .

4.5 Solved Problem 5 (Francis)

H = 60 m, N = 300 rpm, D1 = 1.2 m, B1 = 0.15 m, ηh = 90%, Kf = 0.2, radial discharge. Find guide vane angle, inlet blade angle, Q and power.

- $u1 = \pi \times 1.2 \times 300 / 60 =$ **18.85 m/s** . $\eta h = Vw1\ u1 / gH$, so $Vw1 = 0.9 \times 9.81 \times 60 / 18.85$ = **28.1 m/s** .
- $Vf1 = 0.2\sqrt{(2 \times 9.81 \times 60)} = 0.2 \times 34.31 =$ **6.86 m/s** .

$\alpha = \tan^{-1}(6.86 / 28.1) = 13.7°$; $\theta = \tan^{-1}[6.86 / (28.1 - 18.85)] = \tan^{-1} 0.742 = 36.6°$

$Q = \pi D1 B1 Vf1 = \pi \times 1.2 \times 0.15 \times 6.86 = 3.88\ m^3/s$

Power = ρ Q Vw1 u1 = 1000 x 3.88 x 28.1 x 18.85 = 2.06 MW (water power 2.28 MW, ηh = 90%)

- $Ns = 300\sqrt{2055} / 60^{\wedge}1.25 = 300 \times 45.3 / 167 =$ **81** (medium-speed Francis).

5. Kaplan and Propeller Turbines (Axial Flow)

- Water flows **axially** through the runner; the runner has **3 to 8 blades** on a hub (boss). **Propeller** has fixed blades; **Kaplan** has **adjustable blades** (and adjustable guide vanes: **double regulation**), so efficiency stays high over wide load variation.
- Suited for **low head (5 to 70 m) and large discharge** , Ns = 300 to 1000. Boss diameter Db = 0.35 to 0.6 D0.

$Q = (\pi/4)(D0^2 - Db^2)\, Vf1$; $u1 = u2 = \pi\, D0\, N / 60$ (vane velocity same at inlet and outlet)

- Inlet and outlet velocity triangles are drawn at the mean diameter (Dm); blades twisted along their length because u varies with radius.

Turbine	Head (m)	Ns	Flow	Regulation
Pelton	250 to 2000	10 to 60	Tangential	Spear valve, deflector
Francis	60 to 250	60 to 300	Mixed / radial	Guide vanes
Kaplan	5 to 70	300 to 1000	Axial	Guide vanes + runner blades

6. Draft Tube

6.1 Purpose

- A **gradually expanding tube** connecting the runner outlet to the tailrace; it (i) allows the runner to be placed **above the tailwater level** for easy inspection without loss of head, and (ii) **converts the large kinetic energy at the runner outlet into pressure**, thereby creating a **suction (negative) pressure** at the runner outlet which increases the effective head.

6.2 Theory

- Apply the energy equation between the runner outlet (2) and tail race surface, with runner outlet at height Hs above tailwater and V3 = exit velocity from tube:

$$p2/\gamma = pa/\gamma - Hs - (V2^2 - V3^2)/2g + hf$$

- The pressure at runner outlet is **below atmospheric**; Hs must be limited (cavitation). Draft tube efficiency:

$$\eta d = [\, V2^2/2g - V3^2/2g - hf \,] / (\, V2^2/2g \,) \quad \text{(typically 70 to 90\%)}$$

6.3 Types

- **Conical (straight divergent)**: cone angle 8° to 10° (to prevent flow separation), ηd about 90%; **simple elbow**; **elbow with varying cross-section** (rectangular at outlet); **moody spreading tube** (with a central cone-type pier).

Problem 6: Hs = 4 m, V2 = 6 m/s, V3 = 1.5 m/s, hf = 0.2 m, pa/γ = 10.33 m. $V2^2/2g$ = 1.835 m; $V3^2/2g$ = 0.115 m.

$p2/\gamma = 10.33 - 4 - (1.835 - 0.115) + 0.2 = 4.81$ m (absolute), i.e. 5.5 m below atmospheric; $\eta d = (1.835 - 0.115 - 0.2) / 1.835 = 82.8\%$

7. Unit Quantities, Governing and Characteristic Curves

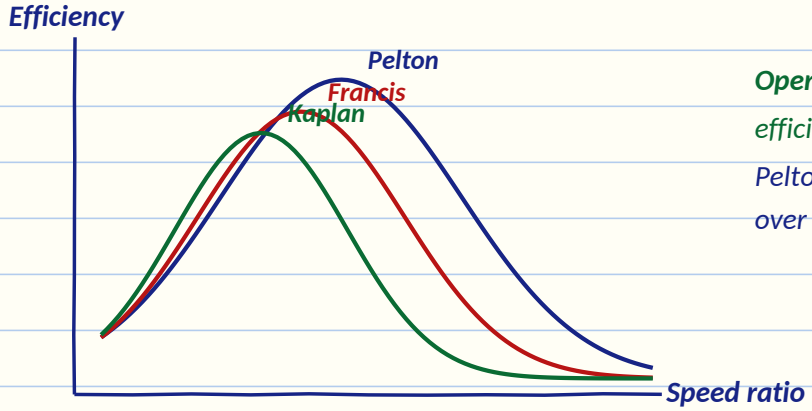
7.1 Unit quantities (head of 1 m)

Unit speed Nu = N / √H ; Unit discharge Qu = Q / √H ; Unit power Pu = P / H^(3/2)

- Used to predict the performance of the **same turbine under different heads** and of model-prototype.

Specific speed Ns = N √P / H^(5/4) (speed of a geometrically similar turbine developing 1 kW under 1 m head)

7.2 Characteristic curves



Operating (constant head) curves:
efficiency vs speed ratio / load.
Pelton efficiency stays high over a wide range of load.

Curve	Plot	Constant
Main (constant head)	Q, P and η against N	Head H
Operating (constant speed)	P, η, Q against load (gate opening)	Speed N
Constant efficiency (Muschel / iso-efficiency)	Contours of η on N-Q plane	η

- **Pelton** : efficiency stays high from 30% to 100% of load; **Francis** : drops at part load; **Kaplan** : nearly flat (adjustable blades); **propeller**: falls sharply at part load.

7.3 Governing of turbines

- **Governing** = keeping speed constant when load changes. Pelton: spear valve (and deflector); Francis: guide vane opening; Kaplan: guide vanes and blade pitch. Controlled by an oil pressure servomotor with a **centrifugal governor**.

8. Cavitation

8.1 Meaning

Cavitation is the **formation, growth and collapse of vapour bubbles** in a flowing liquid at places where the local absolute pressure falls to the **vapour pressure**. When the bubbles move to a high-pressure zone they collapse violently, producing shock waves.

8.2 Causes

- Low pressure at the **runner outlet, blade suction side or draft tube inlet**; **high runner speed or high velocity**; **excessive suction height Hs** (runner too high above tailwater); **high water temperature** (higher vapour pressure); high elevation (lower atmospheric pressure); rough surfaces and sudden changes in flow direction.

8.3 Effects

- **Pitting and erosion** of blade surface (pressures of thousands of atmospheres on the collapse of bubbles); **noise and vibration**; **drop in efficiency and output**; **damage to the runner and draft tube**; sudden failure in extreme cases.

8.4 Thoma's cavitation parameter

$\sigma = (Ha - Hv - Hs) / H$ (Ha = atmospheric head, Hv = vapour pressure head, Hs = runner height above tailwater)

- Cavitation is avoided if $\sigma > \sigma c$ (critical value from tests; larger for higher specific speed). Hence **safe** $Hs = Ha - Hv - \sigma c\, H$.

Problem 7: H = 60 m, σc = 0.10, Ha = 10.3 m, Hv = 0.3 m. $Hs = 10.3 - 0.3 - 0.10 \times 60 =$ **4.0 m** (the runner must not be placed more than 4 m above the tailwater; or set it lower).

8.5 Prevention

- Install the runner **below** (or at a limited height above) tailwater level; use a **draft tube** with correct angle; **smooth, polished** surfaces and good blade profile; **cavitation-resistant materials** (stainless steel, hard-faced welding, bronze); **air admission** (aeration) to the flow; operate within recommended load range.

9. More Solved Problems and Concepts

Problem 8: Series of curved vanes with outlet angle

A jet (d = 75 mm, V = 30 m/s) strikes a series of curved vanes moving at u = 15 m/s. The vane outlet angle is φ = 30° (smooth vanes, k = 1). Find force, power and efficiency.

- $a = 4.418 \times 10^{-3}$ m²; jet KE rate = $0.5 \times 1000 \times 4.418 \times 10^{-3} \times 30^3 = 59.6$ kW.

$F = \rho a V (V - u)(1 + \cos\varphi) = 1000 \times 4.418 \times 10^{-3} \times 30 \times 15 \times 1.866 = 3710$ N

Power = 3710 x 15 = 55.7 kW; $\eta = 2u(V - u)(1 + \cos\varphi)/V^2 = 2 \times 15 \times 15 \times 1.866 / 900 = 93.3\%$

Problem 9: Design of a multi-jet Pelton

P = 6000 kW, H = 500 m, N = 500 rpm, ηo = 85%, Cv = 0.98, speed ratio 0.46, two jets. Find Q, jet diameter, wheel diameter, number of buckets and Ns.

- $Q = P / (\rho g H \eta o) = 6 \times 10^6 / (1000 \times 9.81 \times 500 \times 0.85) =$ **1.44 m³/s**.
- $V1 = 0.98 \sqrt{(2 \times 9.81 \times 500)} = 97.1$ m/s; Q per jet = 0.72 m³/s, $a = 0.72/97.1 = 7.41 \times 10^{-3}$ m², so **d = 97 mm**.

u = 0.46 x 99.05 = 45.6 m/s; D = 60 u / π N = 60 x 45.6 / (π x 500) = 1.74 m (D/d = 17.9)

Z = 15 + D/2d = 15 + 9 = 24 buckets; Ns = 500 √6000 / 500^1.25 = 16.4 (per wheel)

Problem 10: Kaplan turbine

H = 25 m, N = 150 rpm, runner D0 = 3.5 m, hub Db = 1.4 m, Kf = 0.5, ηh = 90%. Find Q, blade angle, guide vane angle, power and Ns.

- $Vf1 = 0.5 \sqrt{(2 \times 9.81 \times 25)} = 11.07$ m/s; $Q = (\pi/4)(3.5^2 - 1.4^2) \times 11.07 = 8.08 \times 11.07 =$ **89.5 m³/s**.
- Mean diameter Dm = 2.45 m; $u = \pi \times 2.45 \times 150 / 60 =$ **19.24 m/s**; $Vw1 = \eta h\, g\, H / u = 0.9 \times 9.81 \times 25 / 19.24 = 11.47$ m/s.

Guide vane angle $\alpha = \tan^{-1}(11.07/11.47) = 44°$; inlet blade angle $\theta = 180° - \tan^{-1}[11.07 / (19.24 - 11.47)] = 180° - 54.9° = 125°$

Power = ρ Q Vw1 u = 1000 x 89.5 x 11.47 x 19.24 = 19.7 MW; Ns = 150 √19740 / 25^1.25 = 377 (Kaplan range)

Problem 11: Unit quantities

A turbine at N = 400 rpm under H = 80 m gives Q = 3 m³/s and P = 2000 kW. Find its performance under H = 60 m.

- $Nu = 400/\sqrt{80} = 44.7$; $Qu = 3/8.944 = 0.335$; $Pu = 2000 / 80^{1.5} = 2.795$.

At 60 m: N = 44.7 x 7.746 = 346 rpm; Q = 0.335 x 7.746 = 2.60 m³/s; P = 2.795 x 60^1.5 = 1299 kW

Runaway speed

- Speed attained by the turbine when the **external load is thrown off** and the governor fails (gate fully open): Pelton about 1.8 to 1.9 N; Francis 2.0 to 2.2 N; Kaplan 2.5 to 3.0 N. The generator and runner must be designed to withstand it.

Selection of a turbine (summary)

Site condition	Choice
Very high head (> 300 m), small discharge	Pelton

Medium head (60 to 250 m), moderate discharge	Francis
Low head (< 60 m), large discharge	Kaplan / propeller
Wide variation in load or head	Kaplan (adjustable blades) or Pelton

Hydro-electric power plant layout



- Components: dam and reservoir, intake with trash rack, **surge tank** (relieves water hammer), penstock, power house with turbine-generator, draft tube and tail race.

10. Quick Revision and Important Questions

Flat plate: $\rho a V^2$. Curved: $\rho a V^2(1 + \cos\theta)$. Moving wheel: $\eta max = 50\%$ at $u = V/2$ (flat); $(1 + \cos\theta)/2$ (curved). Pelton $\eta h = 2u(V1 - u)(1 + k \cos\varphi)/V1^2$. Francis $\eta h = Vw1u1/gH$. $Ns = N\sqrt{P}/H^{(5/4)}$. $\sigma = (Ha - Hv - Hs)/H$.

- Derive the force exerted by a jet on a moving curved vane. Show that the max efficiency for a series of vanes is when $u = V/2$. (10 M)
- Classify hydraulic turbines. Differentiate impulse and reaction turbines. (5 M)
- Explain the working of a Pelton wheel with sketch. Derive its efficiency and explain its design. (10 M)
- Explain the construction and working of a Francis turbine; derive the work done and efficiency. (10 M)
- What is a draft tube? Explain its types and theory. (5 M)
- Explain the characteristic curves and unit quantities. What is specific speed? (5 M)
- What is cavitation? Explain its causes, effects and prevention in hydraulic turbines. (5 M)